

Web Development

Laboratory Manual — Programs

VSE-281-COM

Second Year Computer Engineering (2024 Pattern)

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Experiment No. 1

Title: Simple HTML Page Displaying Personal Details Using Text Formatting Tags

Aim: Create a simple HTML page displaying personal details using text formatting tags.

```
<!DOCTYPE html>
<html>
<head>
  <title>Personal Details</title>
</head>
<body>

  <h1>My Personal Details</h1>

  <p><b>Name:</b> Shreyas Patil</p>
  <p><i>Email:</i> shreyas@example.com</p>
  <p><u>Address:</u> Pune, Maharashtra</p>
  <strong>Hobbies:</strong> Coding, Cricket, Music

</body>
</html>
```

Experiment No. 2

Title: Web Page with Image, Hyperlink and Nested Table

Aim: Design a web page that includes an image, hyperlink, and a nested table.

```
<!DOCTYPE html>
<html>
<head>
    <title>Student Profile Page</title>
</head>
<body>

    <h1>Student Profile Page</h1>

    <h3>Profile Picture:</h3>
    

    <hr>

    <h3>Important Link:</h3>
    <p>Visit the official college website:
        <a href="https://www.unipune.ac.in" target="_blank">Click Here</a>
    </p>

    <hr>

    <h3>Academic Record (Nested Table):</h3>
    <table border="1" cellpadding="10">
        <tr>
            <th>Year</th>
            <th>Semester Details</th>
        </tr>
        <tr>
            <td>First Year</td>
            <td>
                <table border="1">
                    <tr><th>Semester</th><th>GPA</th></tr>
                    <tr><td>Sem I</td><td>8.5</td></tr>
                    <tr><td>Sem II</td><td>8.8</td></tr>
                </table>
            </td>
        </tr>
    </table>

</body>
</html>
```

Experiment No. 3

Title: HTML Registration Form with Name, Email, Gender, DOB and Submit Button

Aim: Create a form with fields: name, email, gender, date of birth, and submit button.

```
<!DOCTYPE html>
<html>
<head>
  <title>Student Registration Form</title>
</head>
<body>

  <h2>Student Registration Form</h2>

  <form>

    <label for="name">Name:</label><br>
    <input type="text" id="name" name="name"
      placeholder="Enter your full name" required>
    <br><br>

    <label for="email">Email ID:</label><br>
    <input type="email" id="email" name="user_email"
      placeholder="example@mail.com" required>
    <br><br>

    <label>Gender:</label><br>
    <input type="radio" name="gender" value="male"> Male
    <input type="radio" name="gender" value="female"> Female
    <input type="radio" name="gender" value="other"> Other
    <br><br>

    <label for="dob">Date of Birth (DOB):</label><br>
    <input type="date" id="dob" name="dob">
    <br><br>

    <input type="submit" value="Submit Form">

  </form>

</body>
</html>
```

Experiment No. 4

Title: Apply Internal, External and Inline CSS to Style a Web Page

Aim: Apply internal, external, and inline CSS to style a web page with headings and tables.

```
<!DOCTYPE html>
<html>
<head>
    <title>CSS Types</title>

    <!-- External CSS -->
    <link rel="stylesheet" type="text/css" href="style.css">

    <!-- Internal CSS -->
    <style>
        .box {
            border: 2px solid black;
            padding: 20px;
            margin: 10px;
        }
        p {
            color: green;
        }
    </style>
</head>
<body>

    <h2>Demonstrating CSS Types</h2>

    <div class="box">

        <!-- Inline CSS -->
        <h3 style="color: red; background-color: yellow;">
            This Heading uses Inline CSS
        </h3>

        <p>This paragraph is styled using Internal CSS.</p>

        <p>This page background is styled using External CSS.</p>

    </div>

</body>
</html>
```

Experiment No. 5

Title: Responsive Web Page using Bootstrap Grid and Components

Aim: Develop a responsive web page using Bootstrap Grid and Components.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Bootstrap Responsive Page</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/
bootstrap.min.css" rel="stylesheet">
</head>
<body>

  <nav class="navbar navbar-dark bg-primary">
    <div class="container-fluid">
      <span class="navbar-brand mb-0 h1">Bootstrap Project</span>
    </div>
  </nav>

  <div class="container mt-5">

    <div class="p-5 mb-4 bg-light rounded-3">
      <h1 class="display-5 fw-bold">Responsive Layout</h1>
      <p class="col-md-8 fs-4">This page uses the Bootstrap Grid system.
      Resize the browser to see the columns stack!</p>
    </div>

    <div class="row text-center">
      <div class="col-md-4 mb-3">
        <div class="card p-3 shadow-sm">
          <h3>Column 1</h3>
          <p>On desktop, 1/3 width. On mobile, full width.</p>
        </div>
      </div>
      <div class="col-md-4 mb-3">
        <div class="card p-3 shadow-sm">
          <h3>Column 2</h3>
          <p>col-md-4 spans 4 out of 12 columns.</p>
        </div>
      </div>
      <div class="col-md-4 mb-3">
        <div class="card p-3 shadow-sm">
          <h3>Column 3</h3>
          <p>Bootstrap makes development faster.</p>
        </div>
      </div>
    </div>

    <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/
bootstrap.bundle.min.js"></script>
  </body>
</html>
```

Experiment No. 6

Title: JavaScript Form Validation (Email and Empty Fields)

Aim: Write a JavaScript program to validate form inputs such as empty fields and email.

```
<!DOCTYPE html>
<html>
<head>
  <title>JS Form Validation</title>
  <script>
    function validateForm(event) {

      var name = document.getElementById("name").value;
      var email = document.getElementById("email").value;
      var msg = "";

      if (name == "") {
        msg += "Name must be filled out.\n";
      }

      var pattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
      if (!pattern.test(email)) {
        msg += "Please enter a valid email address.\n";
      }

      if (msg != "") {
        alert(msg);
        event.preventDefault();
        return false;
      }

      alert("Form submitted successfully!");
      return true;
    }
  </script>
</head>
<body>

  <h2>User Validation Form</h2>

  <form onsubmit="validateForm(event)">

    <label>Full Name:</label><br>
    <input type="text" id="name" name="name">
    <br><br>

    <label>Email Address:</label><br>
    <input type="text" id="email" name="email">
    <br><br>

    <input type="submit" value="Submit Data">

  </form>

</body>
</html>
```

Experiment No. 7

Title: Dynamic Web Page Using JavaScript and DOM Manipulation

Aim: Create a web page that uses JavaScript to display dynamic content using DOM.

```
<!DOCTYPE html>
<html>
<head>
  <title>DOM Manipulation</title>
  <style>
    #myBox {
      width: 300px;
      height: 100px;
      border: 2px solid black;
      display: flex;
      align-items: center;
      justify-content: center;
      margin-top: 20px;
      font-weight: bold;
    }
  </style>
</head>
<body>

  <h1 id="mainHeading">Original Heading</h1>

  <div id="myBox">Status: Normal</div>

  <br>
  <button onclick="changeContent()">Change Content</button>
  <button onclick="changeStyle()">Change Style</button>
  <button onclick="resetPage()">Reset</button>

  <script>
    function changeContent() {
      document.getElementById("mainHeading").innerHTML =
        "Content Updated via DOM!";
      document.getElementById("myBox").innerHTML =
        "Status: Content Changed";
    }

    function changeStyle() {
      var box = document.getElementById("myBox");
      box.style.backgroundColor = "lightblue";
      box.style.color = "darkblue";
      box.style.border = "2px dashed darkblue";
      box.style.borderRadius = "20px";
    }

    function resetPage() {
      document.getElementById("mainHeading").innerHTML =
        "Original Heading";
      var box = document.getElementById("myBox");
      box.innerHTML = "Status: Normal";
      box.style.backgroundColor = "transparent";
      box.style.color = "black";
      box.style.border = "2px solid black";
      box.style.borderRadius = "0px";
    }
  </script>
</body>
```


</html>

Experiment No. 8

Title: Simple Calculator Using JavaScript Functions and Switch-Case

Aim: Write a JavaScript program for a simple calculator using functions and switch-case.

```
<!DOCTYPE html>
<html>
<head>
  <title>JS Calculator</title>
  <style>
    .box { border: 2px solid #333; padding: 20px; width: 280px; }
    input { width: 90%; margin-bottom: 10px; padding: 5px; }
    button { width: 45%; padding: 8px; margin: 2px; cursor: pointer; }
    #result { font-weight: bold; color: blue; margin-top: 10px; }
  </style>
</head>
<body>

  <div class="box">
    <h3>Simple Calculator</h3>

    <input type="number" id="num1" placeholder="Enter first number">
    <input type="number" id="num2" placeholder="Enter second number">

    <button onclick="calculate('+')>Add (+)</button>
    <button onclick="calculate('-')>Sub (-)</button>
    <button onclick="calculate('*')>Mul (*)</button>
    <button onclick="calculate('/')>Div (/)</button>

    <div id="result">Result: </div>
  </div>

  <script>
    function calculate(op) {
      var n1 = parseFloat(document.getElementById("num1").value);
      var n2 = parseFloat(document.getElementById("num2").value);
      var res = 0;

      if (isNaN(n1) || isNaN(n2)) {
        document.getElementById("result").innerHTML =
          "Error: Enter valid numbers";
        return;
      }

      switch(op) {
        case '+': res = n1 + n2; break;
        case '-': res = n1 - n2; break;
        case '*': res = n1 * n2; break;
        case '/':
          if (n2 == 0) {
            document.getElementById("result").innerHTML =
              "Error: Cannot divide by 0";
            return;
          }
          res = n1 / n2;
          break;
      }

      document.getElementById("result").innerHTML = "Result: " + res;
    }
  </script>
```

```
</body>  
</html>
```

Experiment No. 9

Title: PHP Script to Display Welcome Message and Variables

Aim: Design a PHP script to display a welcome message, variables and basic arithmetic.

```
<!DOCTYPE html>
<html>
<head>
    <title>PHP Basics</title>
</head>
<body>

    <h2>PHP Basics</h2>

    <?php
        echo "<b>Hello World!</b><br><br>";

        $name = "Yash";
        $age  = 20;

        echo "Student Name: " . $name . "<br>";
        echo "Student Age: " . $age . "<br>";

        $num1 = 10;
        $num2 = 15;
        echo "Total: " . ($num1 + $num2);
    ?>

</body>
</html>
```

Experiment No. 10

Title: PHP Program to Accept Form Input and Display Using POST Method

Aim: Write a PHP program to accept form input and display it using the POST method.

```
<!DOCTYPE html>
<html>
<head>
    <title>PHP POST Method</title>
</head>
<body>

    <h2>Login Form</h2>

    <form method="POST">
        Username: <input type="text" name="uname" required><br><br>
        Password: <input type="password" name="upass" required><br><br>
        <input type="submit" name="submitBtn" value="Login">
    </form>

    <hr>

    <?php
        if (isset($_POST['submitBtn'])) {
            $username = htmlspecialchars($_POST['uname']);
            echo "<h3>Welcome, " . $username . "!</h3>";
            echo "<p>Data submitted securely using POST method.</p>";
        }
    ?>

</body>
</html>
```

Experiment No. 11

Title: PHP Program for String Manipulation and Arrays

Aim: Implement a PHP program for string manipulation (length, uppercase, reverse) and arrays.

```
<!DOCTYPE html>
<html>
<head>
    <title>String Manipulation</title>
</head>
<body>

    <h2>Strings and Arrays</h2>

    <?php
        // String Manipulation
        $name = "Radha";
        echo "Original: " . $name . "<br>";
        echo "Length: " . strlen($name) . "<br>";
        echo "Uppercase: " . strtoupper($name) . "<br>";
        echo "Reversed: " . strrev($name) . "<br><br>";

        // Array
        $subjects = array("HTML", "CSS", "PHP");
        echo "Number of subjects: " . count($subjects) . "<br>";
        echo "Subjects list: " . implode(", ", $subjects) . "<br>";
    ?>

</body>
</html>
```

Experiment No. 12

Title: PHP Script to Store and Display Values in an Array

Aim: Create a PHP script to store and display values in an array using foreach loop.

```
<!DOCTYPE html>
<html>
<head>
    <title>PHP Arrays</title>
</head>
<body>

    <h2>Student Array</h2>

    <?php
        $students = array("Shreyas", "Yash", "Tejas", "Radha", "Krishna");

        echo "<b>Student List:</b><br>";
        foreach ($students as $index => $student) {
            echo ($index + 1) . ". " . $student . "<br>";
        }

        echo "<br>";
        echo "Total Students: " . count($students) . "<br>";
        echo "First Student: " . $students[0] . "<br>";
        echo "Last Student: " . end($students) . "<br>";
    ?>

</body>
</html>
```